WEEK 7 · LECTURE L3 · UNIT III

Auditing Algorithms in HR

What an algorithmic fairness audit involves: the four fifths test, the NIST framework, and documentation standards for models and data.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

- Describe the steps of an algorithmic fairness audit, mapped to the NIST AI RMF functions.
- Compute and interpret the four fifths (adverse impact) ratio correctly.
- Apply SMACTR, model cards and datasheets for datasets.
- Evaluate the limits of audits and documentation as safeguards.

COURSE OUTCOMES

CLO 3 (fairness, transparency and bias).

THE AUDIT

Eight steps, tagged by NIST function

Step	NIST function
1. Define the decision and the groups affected	Map
2. Map every stage where the tool acts	Map
3. Compute selection rates by group	Measure
4. Apply the four fifths test	Measure
5. Check error rates and proxies	Measure
6. Decide: keep, fix, add human review, or stop	Manage
7. Set human review and appeal	Manage
8. Assign ownership and re-audit dates	Govern

THE FOUR FIFTHS TEST · EEOC (1978)

An illustrative calculation

	Female	Male
Applications	1,000	1,000
Offers	80	140
Selection rate (offers ÷ applications)	8.0%	14.0%
Impact ratio (lower ÷ higher rate)	0.57	—

EXAM LENS

Illustrative figures. A ratio below 0.80 signals adverse impact. Compare selection rates, never raw offer counts, and remember the rule is a rule of thumb: small samples need statistical tests too.

THEORY IN DEPTH · NIST (2023)

Govern, Map, Measure, Manage

THE CORE IDEA

The AI Risk Management Framework organizes AI risk work into four functions.

HOW IT WORKS

Govern sets culture and accountability; Map frames context and risks; Measure assesses them; Manage acts on them.

LIMITS AND CRITIQUE

Voluntary guidance; it says what to do, not how much is enough.

IN PRACTICE

An audit is Measure; without Govern (a named owner) and Manage (a decision), it changes nothing.

EXAM LENS

Name all four functions and show that measurement alone is insufficient.

THEORY IN DEPTH · RAJI ET AL. (2020)

SMACTR: an internal audit in five stages

THE CORE IDEA

Organizations can audit their own algorithms systematically before and after release.

HOW IT WORKS

Scoping, Mapping, Artifact collection, Testing, Reflection, each producing documents.

LIMITS AND CRITIQUE

Internal auditors may lack independence from the teams they review.

IN PRACTICE

An employer should demand the vendor's artifacts, such as training data descriptions and test results, before renewing a contract.

EXAM LENS

Contrast internal audit (SMACTR) with external or regulatory audit; note the independence problem.

THEORY IN DEPTH · MITCHELL ET AL. (2019)

Model cards: report how a model performs, by group

THE CORE IDEA

Every model should ship with a short document on its intended use and limits.

HOW IT WORKS

Model cards report performance broken down by group and intersection, plus ethical considerations.

LIMITS AND CRITIQUE

A card is only as honest as its authors; it does not test the model.

IN PRACTICE

Ask an ATS vendor for a model card showing error rates for women and for applicants from smaller towns.

EXAM LENS

Documentation enables accountability but does not replace testing.

THEORY IN DEPTH · GEBRU ET AL. (2021)

Datasheets: document where training data came from

THE CORE IDEA

Like electronic components, datasets should come with a datasheet.

HOW IT WORKS

It records why the data were collected, how, by whom, who is represented, and recommended uses.

LIMITS AND CRITIQUE

Documentation takes effort and rarely happens without a requirement.

IN PRACTICE

If a hiring model learned from past hires that were mostly male, a datasheet would have revealed it.

EXAM LENS

Link datasheets to Barocas and Selbst's "labeled examples" route to bias.

PRACTICE FOR THE END-TERM EXAMINATION

Questions to rehearse

Design an algorithmic fairness audit for a hiring tool used by a large Indian employer, mapping each step to the NIST AI RMF. **20 marks**

A strong answer: gives ordered steps; correct four fifths method; error and proxy checks; human review; governance; documentation demands.

"Documentation is not accountability." Evaluate model cards, datasheets and internal audits as safeguards. **20 marks**

A strong answer: explains each; notes honesty and independence limits; argues they enable but do not ensure accountability.

Calculate an impact ratio from given selection data and interpret it. **10 marks**

A strong answer: uses rates, not counts; compares with 0.80; notes sample size caution.

CARRY THESE INTO THE EXAMINATION

Three things to remember

1. **An audit is Measure; it needs Govern and Manage to matter**
2. **Compare selection rates, never raw counts**
3. **Demand documentation from vendors before you trust a model**

SET READINGS · FROM THE LESSON PLAN

Read for this lecture

- EEOC Uniform Guidelines on Employee Selection Procedures (29 CFR 1607).
- NIST (2023). AI Risk Management Framework 1.0.

COMPANION

Each set reading is explained in the matching Unit Reader on rksingh.tech/materials.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

Works named on the slides of this lecture. They are background sources, not set readings.

Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Daumé III, H., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM, 64*(12), 86–92. https://doi.org/10.1145/3458723

Mitchell, M., Wu, S., Zaldivar, A., Barnes, P., Vasserman, L., Hutchinson, B., Spitzer, E., Raji, I. D., & Gebru, T. (2019). Model cards for model reporting. In *Proceedings of the Conference on Fairness, Accountability, and Transparency* (pp. 220–229). ACM. https://doi.org/10.1145/3287560.3287596

National Institute of Standards and Technology. (2023). *Artificial intelligence risk management framework (AI RMF 1.0)* (NIST AI 100-1). U.S. Department of Commerce. https://doi.org/10.6028/NIST.AI.100-1

Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., Smith-Loud, J., Theron, D., & Barnes, P. (2020). Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (pp. 33–44). ACM. https://doi.org/10.1145/3351095.3372873

U.S. Equal Employment Opportunity Commission, Civil Service Commission, Department of Labor, & Department of Justice. (1978). Uniform guidelines on employee selection procedures. *Code of Federal Regulations*, Title 29, Part 1607.